

Project 3 - Data Analysis on Housing Prices

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1 Section 0: Methodology and Statistical Theory

The provided implementation performs a multi-faceted statistical analysis on the `USA_Housing.csv` dataset. The methodology is divided into three primary domains:

1.1 Inferential Statistics: T-Testing

We employ a One-Sample T-test to determine if the sample mean ($\bar{x}$) of "Avg. Area Income" significantly deviates from a theoretical population mean ($\mu_0 = 70,000$).

- **Two-Sided Test:** Tests the null hypothesis $H_0 : \mu = 70,000$ against $H_a : \mu \neq 70,000$.
- **One-Sided Tests:** Evaluates directionality (Greater/Less than) to determine if the area is specifically wealthier or poorer than the benchmark.

1.2 Categorical Analysis: Chi-Square Test

To analyze the relationship between income levels and house prices, we discretize continuous data into a 2×2 contingency table using median splits. We then apply the **Chi-Square Test of Independence**:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Where O_i is the observed frequency and E_i is the expected frequency. This determines if "High Income" status is statistically correlated with "High Price" status.

1.3 Predictive Modeling: Simple Linear Regression

The code models the relationship between Income (Independent Variable) and Price (Dependent Variable) using the Ordinary Least Squares (OLS) method:

$$Price = \beta_0 + \beta_1(Income) + \epsilon$$

The strength of this model is quantified via the R^2 coefficient, representing the proportion of variance in price explained by income.

2 Results:

2.1 Hypothesis Testing:

Using the in-built features for the testing process for both the one sided and two sided hypothesis testing cases. We see that in fact, the values we are getting are either 0 or 1. This is actually not unexpected for a large dataset. Here, since the real-world data that we have considered has a size of 5000, even a tiny fluctuation would lead to an incredibly small p-value. Thus, we expect this sort of polarizing value of 0 and 1.

```
--- Hypothesis Testing (Income vs $70,000) ---  
Two-Sided P-value: 0.0000000000  
Right-Tail P-value (Mean > 70k): 1.0000000000  
Left-Tail P-value (Mean < 70k): 0.0000000000
```

Figure 1: The Two Tail and One Tail Test results considering $\mu = 70,000$ as the null hypothesis.

In the context of the USA_Housing dataset, hypothesis testing provides evidence needed to make statistically correct conclusions. When testing if the "Avg. Area Income" is \$70,000, the null hypothesis (H_0) assumes that any deviation from this value is a byproduct of random fluctuations in the 5,000 recorded samples. However, because the calculated p -values are effectively 0.0, we reject H_0 , concluding that the actual wealth level in these areas is meaningfully different from the benchmark.

2.2 Contingency Table:

As is standard, we construct the 2×2 contingency table for the variable quantities of price of the house and income of the area. We can see that (again with a expectedly infinitesimal p-value), there is strong evidence that these variables are correlated.

```
--- 2x2 Contingency Table (Income Level vs Price Level) ---  
price_bin  High Price  Low Price  
income_bin  
High Income      1789      711  
Low Income       711      1789  
  
Chi-Square Statistic: 927.9432  
P-value for Independence: 0.0000000000  
Result: Statistically Significant. The variables ARE correlated.
```

Figure 2: Contingency Table - Income $\times$ Price

The Chi-Square test interprets the level of overlap between high-income brackets and high-priced homes as a statistically significant correlation ($p < \alpha$). Given the large sample size of this dataset, these tests offer high statistical power, ensuring that the identified market trend represent the underlying reality of the data.

3 Linear Regression

A very classic example of linear regression is observing that the price of housing has a linear relationship (in a rough sense) with the mean income of that area. To observe this, we can see the line-of-best-fit through the actual data plotting price against income. This linear regression takes the following form:

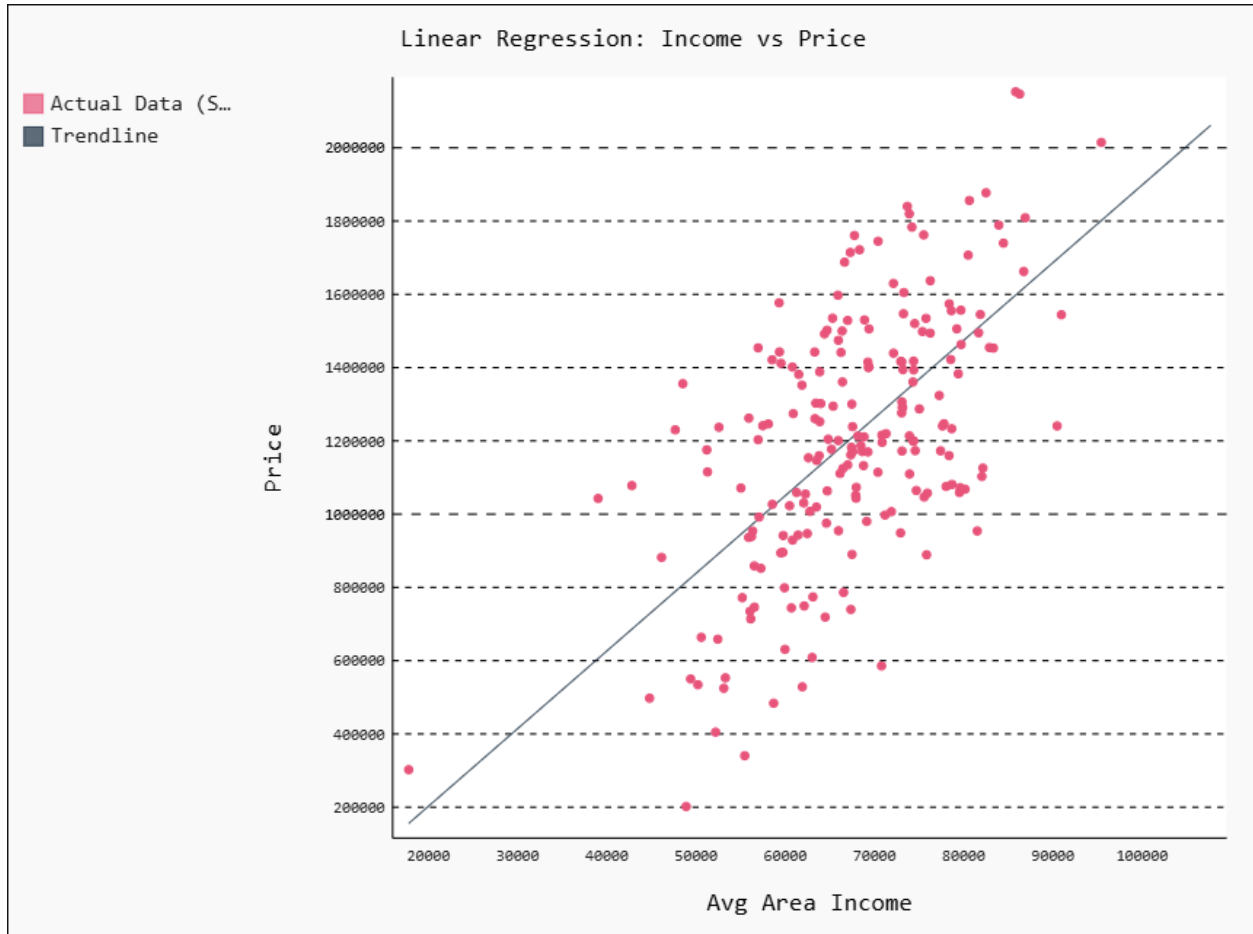


Figure 3: Linear Regression of Price against Income

This shows that we can expect that housing an area will get linearly more expensive as the mean income of that area increases. That is, the rich live in more expensive housing.

4 Implementation and Code Structure

4.1 (a) Data Processing and Statistical Logic

Python Code:

```
import pandas as pd
import numpy as np
import pygal
from scipy import stats
from pygal.style import Style
```

```

# --- 1. DATA LOADING ---
filename = 'USA_Housing.csv'

try:
    df = pd.read_csv(filename)
    # Mapping CSV headers to easier variables
    income = df['Avg. Area Income']
    house_age = df['Avg. Area House Age']
    rooms = df['Avg. Area Number of Rooms']
    bedrooms = df['Avg. Area Number of Bedrooms']
    population = df['Area Population']
    price = df['Price']
    print(f"Dataset loaded: {df.shape[0]} rows found.\n")
except FileNotFoundError:
    print(f"Error: {filename} not found. Ensure it is in the same
          directory.")
    exit()

# --- 2. HYPOTHESIS TESTING (One-Sample T-Test) ---
# Question: Is the average area income significantly different from $70
,000?
null_mean = 70000
t_stat, p_two_sided = stats.ttest_1samp(income, null_mean)

# One-sided tests
# alternative='greater' -> Right Tail
p_right = stats.ttest_1samp(income, null_mean, alternative='greater').
pvalue
# alternative='less' -> Left Tail
p_left = stats.ttest_1samp(income, null_mean, alternative='less').pvalue

print("--- Hypothesis Testing (Income vs $70,000) ---")
print(f"Two-Sided P-value: {p_two_sided:.10f}")
print(f"Right-Tail P-value (Mean > 70k): {p_right:.10f}")
print(f"Left-Tail P-value (Mean < 70k): {p_left:.10f}\n")

# --- 3. 2x2 CONTINGENCY TABLE ---
# Categorizing Income and Price based on their medians to create 2x2 bins
df['income_bin'] = np.where(income > income.median(), 'High Income', 'Low
Income')
df['price_bin'] = np.where(price > price.median(), 'High Price', 'Low
Price')

contingency_table = pd.crosstab(df['income_bin'], df['price_bin'])
chi2, p_chi, dof, expected = stats.chi2_contingency(contingency_table)

print("--- 2x2 Contingency Table (Income Level vs Price Level) ---")
print(contingency_table, "\n")
print(f"\nChi-Square Statistic: {chi2:.4f}")
print(f"P-value for Independence: {p_chi:.10f}")

# Determination of correlation
if p_chi < 0.05:

```

```

        print("Result: Statistically Significant. The variables are correlated
              .\n")
    else:
        print("Result: Not Significant. The variables are independent.\n")

# --- 4. LINEAR REGRESSION ---
# Modeling Price (dependent) based on Area Income (independent)
slope, intercept, r_value, p_reg, std_err = stats.linregress(income, price
)

print("--- Linear Regression Results (Price ~ Income) ---")
print(f"Slope: {slope:.4f}")
print(f"Intercept: {intercept:.4f}")
print(f"R-squared: {r_value**2:.4f}\n")

```

4.2 (b) Visualization Output:

```

# --- 5. VISUALIZATIONS WITH PYGAL ---

# Regression Visualization: Scatter Plot + Trendline
reg_style = Style(colors=('E8537A', '1C3144'))
reg_chart = pygal.XY(
    stroke=False,
    title='Linear Regression: Income vs Price',
    x_title='Avg Area Income',
    y_title='Price',
    style=reg_style
)

# Adding a sample of 200 points for performance/clarity in the SVG
scatter_data = [tuple(x) for x in df[['Avg. Area Income', 'Price']].values
[:200]]
reg_chart.add('Actual Data (Sample)', scatter_data)

# Adding the Regression Line
x_min, x_max = income.min(), income.max()
reg_chart.add('Trendline', [
    (x_min, slope * x_min + intercept),
    (x_max, slope * x_max + intercept)
], stroke=True, show_dots=False)

reg_chart.render_to_file('housing_regression.svg')

# Contingency Table Visualization: Dot Chart
dot_chart = pygal.Dot(title='Contingency Table Frequencies', x_title='
    Price Category', y_title='Income Category')
dot_chart.x_labels = ['High Price', 'Low Price']
dot_chart.add('High Income', [contingency_table.loc['High Income', 'High
    Price'],
                                contingency_table.loc['High Income', 'Low
    Price']])

```

```
dot_chart.add('Low Income', [contingency_table.loc['Low Income', 'High  
Price'],  
                             contingency_table.loc['Low Income', 'Low  
Price']])  
  
dot_chart.render_to_file('contingency_dot_chart.svg')
```

Explanation: The code utilizes `pandas` for vectorized data manipulation, allowing for efficient "binning" of continuous variables. The `scipy.stats` library provides the computational backbone for the p-values, while `pygal` generates scalable vector graphics (SVG) to visualize the regression trendline against the raw data points.